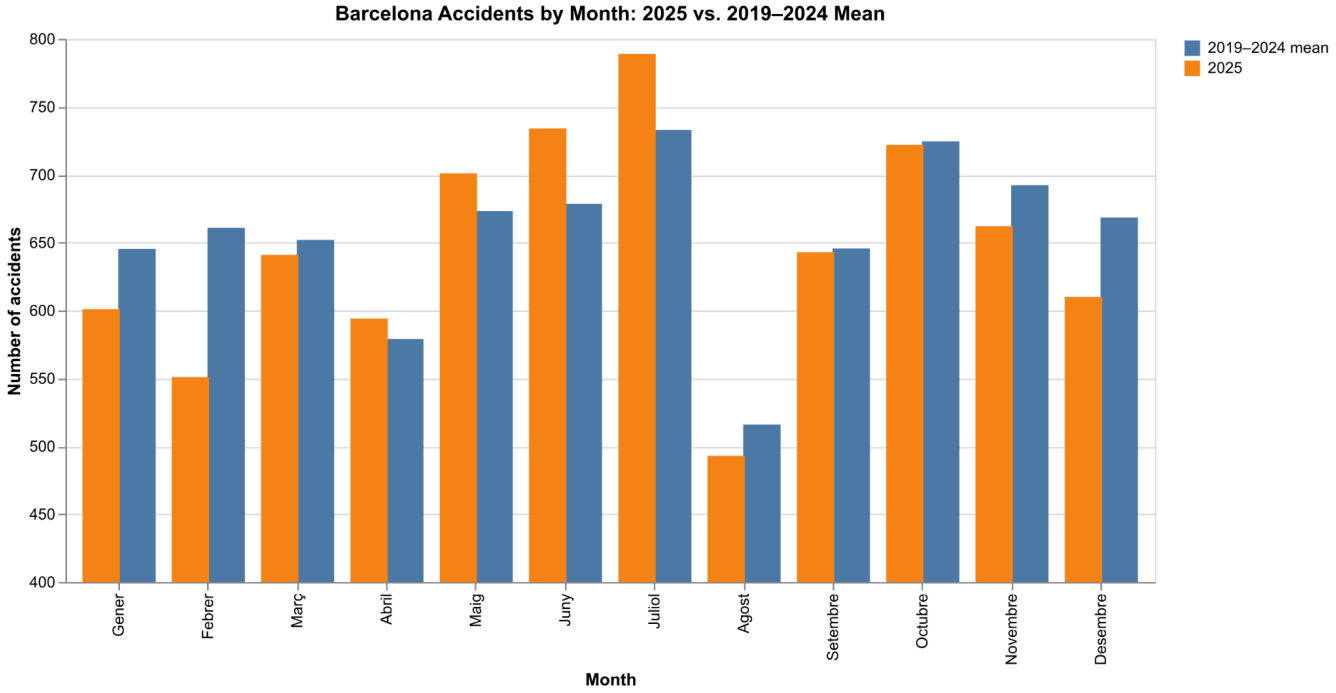
TASK 1.3

Chart 1:

Chart name: Barcelona Accidents by month: 2025 vs. 2019-2024 mean.
Goal: To show when there tends to be more accidents, in the most recent year as well as in the past, so that we know when to implement more controls.

Explain aggregation, precision, data selection decisions:
We aggregated the data by month in order to show the differences in the amount of accidents depending on the time of the year.

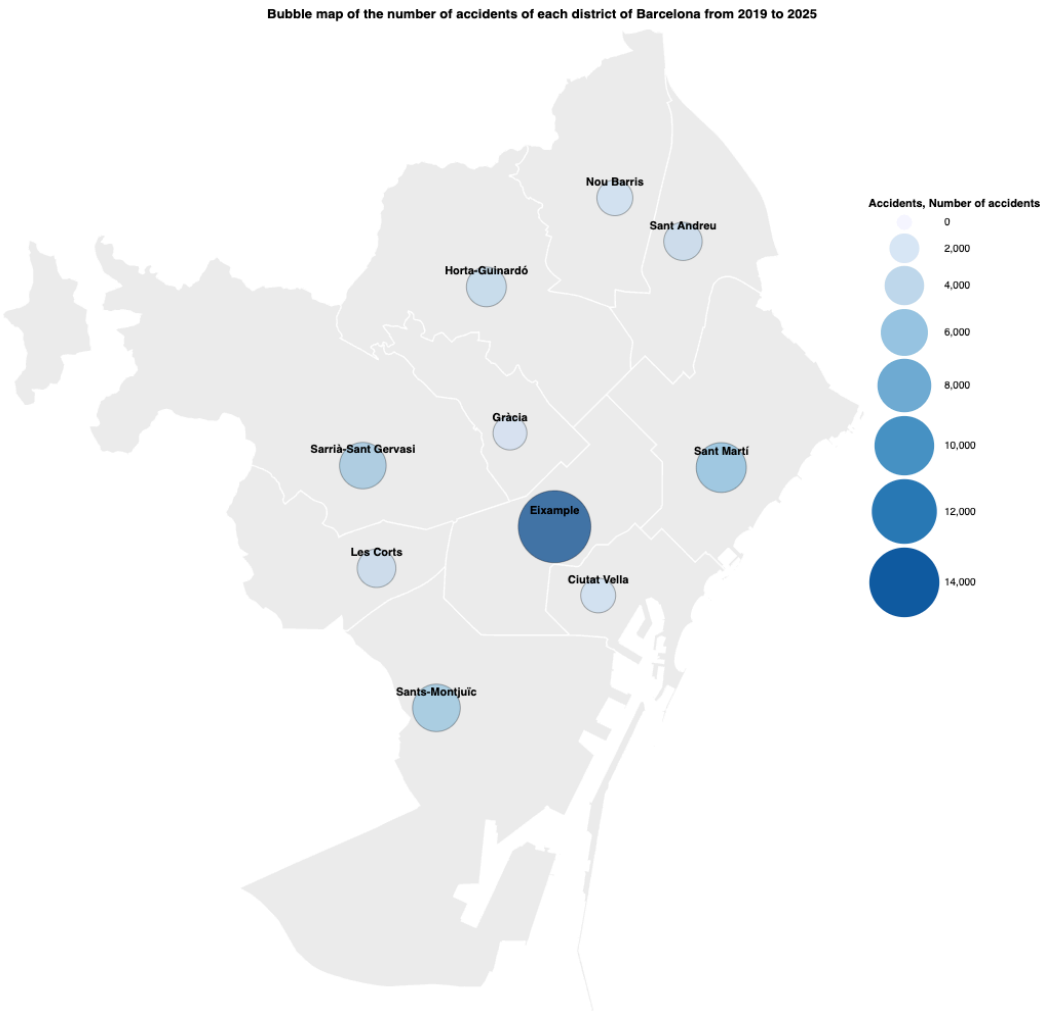


Field	Encoding	Function	Relation with perception theory	Legend
Month	In the x-axis	Show the timeline for each year	As it is separated with two joined bars for each month the user is able to visualize faster the comparison of both of them, depending on the month.	Name of the months in chronological order
Number of accidents	In the y-axis	Show how many accidents happen in each month	We can easily visualize quantity in the y-axis	Amounts starting at 400
Year groups	Colored bars	Show the historical mean along with the most recent information	The bars are together in each month so it is easy to compare and see if there are many notable differences. The colors are quite different so there is no confusion.	Blue -> 2019-2024 orange-> 2025

Chart 2:

Chart name: Bubble map of the number of accidents of each district of Barcelona 2019-2026
Goal: To show which district tends to have more accidents

Explain aggregation, precision, data selection decisions:
Sr. Pere Capdevila wanted to know how to distribute the new budget of DGT for the traffic cameras for each district. So it was very important to show the districts with the most accidents in a visual way.
As we wanted to really differentiate each district we used a map with each district separated and the bubble point is at its center.
We are just giving an approximation of the numbers, as precision is not necessary as we only want to know which districts have more accidents.



Field	Encoding	Function	Relation with perception theory	Legend
District	Latitude, longitude	Show where there are more and less accidents	We can recognize each district in a familiar map	Name of the district on the map
Number of accidents	Bubble size + color opacity	Show how many accidents happened in each district	Our perception is more drawn to the bigger and more intense coloured bubble	The size and opacity of the colored bubble increases as the number of accidents increase